THE HOM FUNCTOR IS LIMIT PRESERVING FOR F OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-hom (canonical tag 08JT). cited at statement line 556. Standing situation: F, G over X -> B.
- Remark remark-hom-base-change (canonical tag 08JW). cited at proof line 571. Base-change reduction to B = Spec(R) affine.
- Lemma spaces-properties-lemma-quasi-compact-affine-cover (canonical tag 03H6). cited at proof lines 577-578. Affine etale covers U -> X and V -> U x_X U.
- Lemma lemma-hom-sheaf-in-X (canonical tag 08K3). cited at proof line 581. Hom(F, G) as an equalizer over the etale cover.
- Lemma algebra-lemma-module-map-property-in-colimit (canonical tag 05LI). cited at proof line 602. Hom commutes with the directed colimit for finitely presented M.

Source reference. Original definition 08JT: sources/stacks-chow/tags.txt, source lines 10861–10861. Cited at statement line 556: the lemma is phrased in this situation; load-bearing.

Source reference. Original remark 08JW: sources/stacks-chow/tags.txt, source lines 10864–10864. Cited at proof line 571: justifies replacing B, X, F, G by their pullbacks to T_0; load-bearing.

Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Cited at proof lines 577-578: the affine etale cover U -> X; load-bearing.

Source reference. Original lemma 08K3: sources/stacks-chow/tags.txt, source lines 10871–10871. Cited at proof line 581: Hom(F, G) is the equalizer of the two maps over U and V; load-bearing.

Source reference. Original lemma 05LI: sources/stacks-chow/tags.txt, source lines 7237–7237. Cited at proof line 602: the colimit Hom map is bijective when M is of finite presentation; load-bearing.

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Lemma 0.2. *In Situation 08JT. If $\mathcal{F}$ is of finite presentation and f is quasi-compact and quasi-separated, then $Hom(\mathcal{F}, \mathcal{G})$ is limit preserving.*

Proof. Let $T = \lim_{i \in I} T_i$ be a directed limit of affine B-schemes. We have to show that

$$Hom(\mathcal{F}, \mathcal{G})(T) = \operatorname{colim} Hom(\mathcal{F}, \mathcal{G})(T_i)$$

Pick $0 \in I$. We may replace B by T_0, X by X_{T_0}, $\mathcal{F}$ by $\mathcal{F}_{T_0}$, $\mathcal{G}$ by $\mathcal{G}_{T_0}$, and I by $\{i \in I \mid i \geq 0\}$. See Remark 08JW. Thus we may assume $B = \operatorname{Spec}(R)$ is affine.

When B is affine, then X is quasi-compact and quasi-separated. Choose a surjective étale morphism $U \to X$ where U is an affine scheme (Properties of Spaces, Lemma 03H6). Since X is quasi-separated, the scheme $U \times_X U$ is quasi-compact and we may choose a surjective étale morphism $V \to U \times_X U$ where V is an affine scheme. Applying Lemma 08K3 we see that $Hom(\mathcal{F}, \mathcal{G})$ is the equalizer of two maps between

$$Hom(\mathcal{F}|_U, \mathcal{G}|_U) \quad \text{and} \quad Hom(\mathcal{F}|_V, \mathcal{G}|_V)$$

This reduces us to the case that X is affine.

In the affine case the statement of the lemma reduces to the following problem: Given a ring map $R \to A$, two A-modules M, N and a directed system of R-algebras $C = \operatorname{colim} C_i$. When is it true that the map

$$\operatorname{colim} \operatorname{Hom}_{A \otimes_R C_i}(M \otimes_R C_i, N \otimes_R C_i) \longrightarrow \operatorname{Hom}_{A \otimes_R C}(M \otimes_R C, N \otimes_R C)$$

is bijective? By Algebra, Lemma 05LI this holds if $M \otimes_R C$ is of finite presentation over $A \otimes_R C$, i.e., when M is of finite presentation over A. □